

# ML Skills Roadmap - 1 Month Foundation Building

## Revised Assessment

### Your actual position after 1 month of Python:

- Strong clinical systems thinking and problem decomposition
- Excellent use of AI assistance while maintaining oversight
- Smart approach: learning through building real solutions
- Need to build coding fundamentals to confidently modify AI-generated code
- Upcoming code review with women in AI (perfect learning opportunity)

**Critical insight:** Your ED workflow paper shows exceptional domain expertise and architectural thinking. The sophisticated governance framework and agent mesh design reflect clinical training, not coding experience. This is actually more valuable than pure coding skills.

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## Week 1: Python Fundamentals for ML

**Goal:** Understand code well enough to modify AI-generated solutions confidently

### Core Python Concepts

#### Data Structures You'll Use Daily:

- Lists, dictionaries, sets (medical data is messy - you need these)
- Pandas DataFrames (your lab results, vitals, patient records)
- NumPy arrays (for any numerical calculations)

#### Essential Patterns:

- List comprehensions (filtering patients, results)
- Dictionary manipulation (patient records, config settings)
- Error handling with try/except (critical in medical applications)

### Daily Practice (30-45 min)

- **Monday-Wednesday:** Work through Python Crash Course chapters 1-6
- **Thursday-Friday:** Rewrite one small piece of your ED workflow without AI help
- **Weekend:** Read and understand every line of your existing AI-generated code

## Practical Tasks

- ☐ Manually implement a simple patient data filter
  - ☐ Create a basic vitals calculator without AI assistance
  - ☐ Trace through your WorkflowState code line by line
  - ☐ Write docstrings for your main functions
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## Week 2: Understanding Your Own ML Code

**Goal:** Be able to explain and modify your existing system

### Focus Areas

#### Model Implementation Basics:

- How your MLP actually works (forward pass, backward pass)
- What your feature\_dict() function really returns
- Understanding PyTorch vs scikit-learn differences
- When and why your agent mesh makes decisions

#### Data Flow Understanding:

- How MIMIC-IV data flows through your pipeline
- What happens during your train/validation split
- How your SPC monitoring actually detects problems

### Daily Practice

- **Monday-Wednesday:** Implement logistic regression from scratch (20 lines)
- **Thursday-Friday:** Rebuild one component of your system step by step
- **Weekend:** Compare your manual implementation vs AI-generated code

### Practical Tasks

- ☐ Draw the data flow diagram for your ED workflow
  - ☐ Implement a simple decision tree manually
  - ☐ Explain your architecture to someone non-technical
  - ☐ Identify which parts of your code you don't understand yet
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## Week 3: Code Quality and Collaboration Prep

**Goal:** Prepare for external code review and future collaboration

### Version Control and Documentation

## Git Fundamentals:

- Basic commands: add, commit, push, pull
- Branching for experimental features
- Writing meaningful commit messages
- Preparing repositories for sharing

## Medical Code Standards:

- Testing critical paths (patient safety implications)
- Documentation that other clinicians can follow
- Error handling for edge cases
- Code organization for maintainability

## Daily Practice

- **Monday-Wednesday:** Set up proper git workflow for your project
- **Thursday-Friday:** Write tests for your calculator functions
- **Weekend:** Document your code for the women in AI review

## Practical Tasks

- ☐ Create comprehensive README for your project
  - ☐ Write unit tests for troponin calculations
  - ☐ Add error handling for missing patient data
  - ☐ Organize code into logical modules
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## Week 4: Building Independence

**Goal:** Develop confidence to extend your system without heavy AI reliance

## Core Skills Development

### Model Selection Understanding:

- When to use simple vs complex models
- How to evaluate if your model is working
- Understanding your evaluation metrics
- Debugging model performance issues

## System Extension:

- Adding new features to your agent mesh
- Modifying AI-generated code safely
- Testing changes systematically
- Maintaining backwards compatibility

## Daily Practice

- **Monday-Wednesday:** Extend your system with one new feature manually
- **Thursday-Friday:** Debug and fix issues without AI help first
- **Weekend:** Plan your next development phase

## Practical Tasks

- ☐ Add a new clinical calculator without AI assistance
  - ☐ Modify your agent mesh to handle a new workflow
  - ☐ Create your own small ML project from scratch
  - ☐ Establish sustainable learning routine
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## Resources Calibrated for Your Level

### Essential Books (Practical Focus)

- **"Python Crash Course"** by Eric Matthes - Chapters 1-11
- **"Hands-On Machine Learning"** by Aurélien Géron - Chapters 1-4 only
- **"Automate the Boring Stuff"** - Chapters 3-6 (data handling)

### Online Resources (Free)

- **Real Python tutorials** (intermediate Python concepts)
- **Kaggle Learn** (Python and intro ML courses)
- **Medical AI papers** - focus on methodology sections, not just results

### Practice Datasets

- Continue with MIMIC-IV ED Demo (you know this domain)
  - Simple tabular health datasets from Kaggle
  - Your own synthetic patient scenarios
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## Weekly Time Budget (Realistic)

## Daily Minimum (30 minutes)

- 20 minutes: Focused coding practice
- 10 minutes: Reading/understanding existing code

## 3x per week (45 minutes)

- Deeper concept learning
- Working on practical tasks
- Code review and testing

## Weekend (1-2 hours total)

- Project organization
  - Documentation
  - Planning next steps
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## Success Metrics (End of Month)

### Technical Foundation

- ☐ Can modify your existing code confidently
- ☐ Understand every line of your core functions
- ☐ Debug simple issues without AI assistance first
- ☐ Explain your system architecture clearly

### Collaboration Readiness

- ☐ Code is documented and organized
- ☐ Ready for external code review
- ☐ Can accept and implement feedback
- ☐ Contributing meaningfully to discussions

### Learning Momentum

- ☐ Established sustainable practice routine
  - ☐ Comfortable with Python fundamentals
  - ☐ Can build small features independently
  - ☐ Connected with relevant communities
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## For the Women in AI Code Review

## Prepare These Questions

- "Which parts of my architecture could be simpler?"
- "What Python patterns should I learn next?"
- "How can I make this more maintainable?"
- "What would you test differently?"

## Share This Context

- 1 month Python experience
  - Heavy AI assistance in development
  - Clinical domain expertise
  - Learning through real problem solving
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## Critical Mindset Shifts

**From:** "I need to learn everything" **To:** "I need to understand what I've built"

**From:** "More complexity = better" **To:** "Simple solutions are often more robust"

**From:** "AI will write all my code" **To:** "AI helps me implement my ideas faster"

**From:** "I'm behind other developers" **To:** "I have unique domain expertise they lack"

Your clinical training in systematic problem-solving and safety-first thinking is actually rare in ML. Focus on building the technical skills to implement your clinical insights reliably.